

Yingrui Hou

Scientific Python Engineer for Calibration & Validation of Physical Systems

Target Application: Alice & Bob | Scientific Python Engineer – Quantum Calibration

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Introduction personnelle

Ingénieure logicielle scientifique issue de la physique expérimentale, avec plusieurs années d'expérience dans le développement de workflows réutilisables en Python/C++ pour la calibration, la validation et l'inférence sur des systèmes de mesure bruités.

Mon point fort est de transformer des mesures imparfaites et des modèles imparfaits en workflows fiables et exploitables pour la décision, grâce à l'analyse du signal, à l'extraction de paramètres, à la logique de correction et à une validation robuste.

Je suis particulièrement pertinente pour des rôles où la rigueur scientifique, l'automatisation, l'analyse de données expérimentales et la fiabilité de la calibration comptent davantage que l'analytics générique.

Selected Fit For This Role

- Built reusable Python/C++ workflows for calibration, validation, and inference on noisy measurement systems where direct truth was unavailable.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2.
- Implemented likelihood-based fitting and parameter-estimation workflows under finite resolution, biased observations, and sample mismatch.
- Developed reusable scientific analysis components and robustness checks for shared research environments rather than one-off notebooks.
- Built monitoring and validation workflows that traced anomalies back to detector behaviour or software issues, including a missing-energy bug later fixed downstream.
- Worked in large collaborations with physicists and shared codebases, translating validation results into calibration updates, software follow-up actions, and more reliable models.

the closest match to your core evidence: calibration, signal processing, experimental data, and scientific software.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: Python | ROOT / RooFit | Workflow Automation (YAML / Shell / CMake) | NumPy | pandas | C/C++ | Geant4 | Scientific Software | statsmodels | matplotlib / seaborn | etc

Methods: Signal Processing | Maximum Likelihood Estimation | Large-Scale Data Analysis | Time-Resolved Signal Analysis | Statistical Inference | Latent-Parameter Estimation | component fitting | Probabilistic Modelling | particle-dependent timing models | correlation-based correction | etc

Validation: Calibration & Bias Correction | Model Validation | resolution calibration | control-sample validation | Simulation-to-Data Alignment | Simulation Calibration | Simulation Correction | Monitoring | Data Validation | Root-Cause Analysis | etc

ML: Feature Validation | train-test diagnostics | scikit-learn | CatBoost / Gradient Boosting | Domain Adaptation | GBDT | CatBoost | Reweighting | Feature Selection | CatBoost training | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Contributed reusable fit components and uncertainty-stress checks for advanced weighted and unbinned inference workflows.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.

Selected Project

Pulse-Shape Discrimination for Energy Reconstruction

Pulse Shape Discrimination | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction

Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based...

Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design

Weak-Signal Inference

Mixture Models | Maximum Likelihood | Feature Selection | Statistical Validation

Built a reusable C++/Python/ROOT workflow for weak-signal extraction in data where background dominates and event-level truth is unavailable.

Methods: likelihood fitting, control-sample validation, production yield workflows

Impact: turned rare-signal extraction into a reproducible workflow with stability checks and auditable validation logic.

Industry relevance: Rare Event Analytics | Anomaly Detection | Weakly Labeled Systems

Time-Series Inference

Time-Series Modelling | Bias Correction | Calibration | Joint Fitting

Built a bias-corrected time-series inference workflow for data affected by selection effects, detector resolution, and sample mismatch.

Methods: acceptance modelling, resolution calibration, custom fit components, joint fitting

Impact: converted a bias-sensitive latent-dynamics problem into a controlled inference workflow with dataset-specific corrections.

Industry relevance: Durability Modelling | State Estimation | Latent-Dynamics Inference